FLIGHT CONTROLLER

ARKV6S

Single-IMU FMUv6X autopilot module on the Pixhawk Autopilot Bus

EXPOSED INTERFACES: CAN | UART | I2C | SPI | PWM | USB | ETH | RC | PPM

The USA-built ARKV6S flight controller is a cost-optimised single-IMU variant of the ARKV6X, based on the FMUv6X and Pixhawk Autopilot Bus (PAB) open source standards. An STM32H743 processor reads one industrial-grade IMU together with a barometer and magnetometer, delivering a complete navigation solution without the multi-IMU redundancy of the ARKV6X.

Conforming to the PAB form factor, it drops into any PAB-compatible carrier — the ARK PAB Carrier, ARK Jetson PAB Carrier or ARK Jetson PAB V3 — so one autopilot module moves between airframes and compute payloads.

01 KEY FEATURES

- Single industrial IMU with independent power domain
  - 1 × TDK InvenSense IIM-42653 (SPI1)
- Bosch BMP390 barometer and ST IIS2MDC magnetometer on internal I2C
- Cypress FM25V02A 256 kbit FRAM for non-volatile parameter storage
- 1 W resistive sensor heater under closed-loop MCU control
- Supercapacitor-backed real-time clock
- microSD logging on a 4-bit SDMMC interface
- Pixhawk Autopilot Bus (PAB) form factor — carrier-independent

02 SPECIFICATIONS

PROCESSOR & MEMORY	
MCU	ST STM32H743IIK6, 176-UFBGA
Flash / RAM	2 MB / 1 MB
FRAM	FM25V02A, 256 kbit, 40 MHz SPI
Removable storage	microSD, SDMMC 4-bit, push-push
Oscillators	16.000 MHz + 32.768 kHz ±20 ppm

ELECTRICAL	
Supply voltage	5 V
Supply current	500 mA max
— main system	300 mA
— sensor heater	200 mA
Operating temperature	−40 °C to +85 °C

MECHANICAL	
Dimensions	36 × 29 × 5 mm
Weight	5.0 g
Mounting	4 × Ø2.3 mm on 31.4 × 24.4 mm

AT A GLANCE	
Standard	FMUv6X / PAB
MCU	STM32H743IIK6
IMUs	1 (industrial)
Baro / Mag	BMP390 / IIS2MDC
Supply	5 V, 500 mA max
Interface	100 + 50 pin PAB
Dimensions	36 × 29 × 5 mm
Weight	5.0 g
Operating temp	−40 to +85 °C
Hardware rev	Rev 1.0

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- FCC compliant
- FMUv6X open standard
- Pixhawk Autopilot Bus DS-010 standard

ORDERING & RESOURCES

STORE arkelectron.com/product/arkv6s-flight-controller/

DOCS docs.arkelectron.com/products/flight-controller/arkv6s

CAD github.com/ARK-Electronics/ark-hardware

03 PIXHAWK AUTOPILOT BUS INTERFACE

PAB CONNECTOR	HIROSE PART	SIGNALS CARRIED
PAB X1 (J2)	DF40C-100DP-0.4V, 100 pos	8 × PWM · CAN1 / CAN2 · 8 × UART (see serial map) · I2C1–I2C3 · USB with VBUS sense · PPM and RC input · SWD · safety switch, buzzer, armed · 3 × power-input sense · ADC · 5 V input
PAB X2 (J3)	DF40C-50DP-0.4V, 50 pos	Ethernet RMII with power enable · external SPI6 (2 × CS, reset, 2 × DRDY) · SPIX_SYNC · 17 reserved pins

04 SENSOR BUS MAP

DEVICE	INTERFACE	ADDRESS / SELECT	FUNCTION
IIM-42653 (U9)	SPI1	nCS1	Industrial 6-axis accel / gyro
BMP390 (U11)	I2C	0x76	Barometric pressure
IIS2MDC (U5)	I2C	0x1E	3-axis magnetometer

05 CARRIER BOARD OPTIONS

ARK CARRIER BOARD	AVIONICS I/O	ADDS
ARK Pixhawk Autopilot Bus Carrier	JST-GH	Full Pixhawk-standard breakout, 100 Mbps Ethernet, dual power
ARK Jetson PAB Carrier	JST-GH	NVIDIA Jetson Orin compute on Pixhawk-standard ports
ARK Jetson PAB V3	Pico-Clasp	Jetson Orin compute, triple power inputs, Ethernet switch
ARK VOXL2 RTK PAB Carrier	Molex	ModalAI VOXL2 compute; Blue UAS Framework listed

06 ALTERNATIVES IN THE ARK AUTOPILOT LINE

AUTOPILOT	INERTIAL SENSING	POWER INPUT	WHEN TO CHOOSE IT
ARKV6X	3 IMUs, voting	5 V via carrier	Redundant sensing, highest reliability
ARKV6X Extended Range	3 IMUs (IIM-42653)	5 V via carrier	Higher angular-rate range
ARK FPV	1 industrial IMU	3–12S direct	Standalone 30.5 mm build, 12 V payload rail

07 SERIAL REFERENCE & FIRMWARE SUPPORT

UART	DEVICE	PORT
USART1	/dev/ttyS0	GPS
USART2	/dev/ttyS1	TELEM3
USART3	/dev/ttyS2	Debug Console
UART4	/dev/ttyS3	UART4 & I2C
UART5	/dev/ttyS4	TELEM2
USART6	/dev/ttyS5	PX4IO/RC
UART7	/dev/ttyS6	TELEM1
UART8	/dev/ttyS7	GPS2

FIRMWARE & SOFTWARE SUPPORT

- PX4 Autopilot — ships by default; ArduPilot supported
- Board-to-board pinout follows the DS-010 Pixhawk Autopilot Bus Standard

> Carrier required. All I/O is presented on the two PAB board-to-board connectors; wiring connectors are provided by the carrier board.

ARK-DS-ARKV6S | Issue 2026-08 | Compiled from the ARKV6S Production schematic Rev 1.0 (29 Jan 2026), the ARKV6S Production bill of materials, the ARKV6S production STEP model, and ARK Electronics published documentation. Specifications are subject to change without notice; confirm against current published documentation before design commitment. ARK Electronics products are designed and manufactured in the USA.